



Medical federated learning with joint graph purification for noisy label learning

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ARTICLE INFO

Keywords:

Medical federated learning
Label noise
Graph purification
Negative learning
Joint optimization

ABSTRACT

In terms of increasing privacy issues, Federated Learning (FL) has received extensive attention in medical imaging. Through collaborative training, FL can produce superior diagnostic models with global knowledge, while preserving private data locally. In practice, medical diagnosis suffers from intra-/inter-observer variability, thus label noise is inevitable in dataset preparation. Different from existing studies on centralized datasets, the label noise problem in FL scenarios confronts more challenges, due to data inaccessibility and even noise heterogeneity. In this work, we propose a federated framework with joint Graph Purification (FedGP) to address the label noise in FL through server and clients collaboration. Specifically, to overcome the impact of label noise on local training, we first devise a noisy graph purification on the client side to generate reliable pseudo labels by progressively expanding the purified graph with topological knowledge. Then, we further propose a graph-guided negative ensemble loss to exploit the topology of the client-side purified graph with robust complementary supervision against label noise. Moreover, to address the FL label noise with data silos, we propose a global centroid aggregation on the server side to produce a robust classifier with global knowledge, which can be optimized collaboratively in the FL framework. Extensive experiments are conducted on endoscopic and pathological images with the comparison under the homogeneous, heterogeneous, and real-world label noise for medical FL. Among these diverse noisy FL settings, our FedGP framework significantly outperforms denoising and noisy FL state-of-the-arts by a large margin. The source code is available at <https://github.com/CUHK-AIM-Group/FedGP>.

1. Introduction

Nowadays, deep learning techniques have been widely studied in computer-aided diagnosis, and provide clinicians with reliable and effective assistance for diagnosis (Liu et al., 2020; Esteva et al., 2017). The great success of these works relies on the supervision knowledge of clean datasets, which are often collected from multiple institutions. However, due to the high risk of privacy leakage, the existing learning paradigm based on large centralized datasets would become difficult and unsustainable. To address the privacy issue, Federated Learning (FL) (Kairouz et al., 2021) provides a decentralized collaborative scheme to train local models at multiple data owners (i.e., clients), where a global server iteratively aggregates these uploaded local models. As such, the aggregated models with global knowledge can outperform local models trained at any single client, without the leakage of raw data. Since FL has significant feasibility in medical

institution collaboration, many studies explored diverse applications and technical improvements (Roth et al., 2020; Andreux et al., 2020).

Current medical FL works follow a common assumption that private medical samples across clients are annotated with clean labels. In practice, accurate clinical diagnosis requires high expertise of physicians, and label noise is often inevitable in manual annotation during dataset preparation (Zhu et al., 2021), due to intra-/inter-observer variability. Essentially, medical samples with noisy labels confuse the supervised training and impede the model to learn discriminative knowledge, thereby severely degrading the performance. Despite the label noise in medical and natural scenarios has been extensively studied with the centralized setting (Liu et al., 2021; Ju et al., 2022), the problem of label noise in medical FL that is closer to practical needs has not been explored. Due to the challenges of data inaccessibility, label noise in FL scenarios becomes more complicated and intractable. Hence, robust FL

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<https://doi.org/10.1016/j.media.2023.102976>

Received 6 October 2022; Received in revised form 8 February 2023; Accepted 18 September 2023

Available online 4 October 2023

1361-8415/© 2023 Published by Elsevier B.V.

frameworks capable of handling label noise are critical to the medical community.

Firstly, medical FL frameworks need a powerful denoising technique to eliminate the impact of label noise on the local training of each client. Since iterative communications between server and clients may accumulate the damage induced by label noise, the local training in FL is more fragile to noisy samples. Most existing studies distinguish noisy labels based on some predefined criteria of an individual sample (e.g., small loss [Gui et al., 2021](#), prediction confidence [Bai and Liu, 2021](#) and contrastive divergence [Wei et al., 2020](#)), but neglect the sample relationship to explore clean samples. In medical imaging, different pathologies for a specific disease are often correlated ([Chen et al., 2022a](#)), and can provide structural knowledge among samples to discover beneficial features against label noise. A rational idea is to select a few but most reliable samples as clean, and progressively correct the rest noisy labels by exploiting the topological knowledge of sample relationship. In addition, negative learning ([Kim et al., 2019](#)) can utilize the rest noisy samples to combat label noise instead of discarding their supervision completely, but randomly selecting one complementary label poses the risk of incorrect and limited information. Therefore, the topological relationship among the purified and noisy samples can facilitate negative learning to complement more reliable supervision for FL optimization.

Moreover, existing denoising techniques targeted at centralized training are limited to the local data of each client, and cannot integrate the global knowledge across decentralized clients to combat label noise. In FL, the server aggregation ([McMahan et al., 2017](#)) may introduce noisy information from other clients and cause diagnostic models to overfit the label noise. In addition, medical institutions with different labeling capabilities ([Zhang et al., 2020](#)) may encounter heterogeneous label noise (e.g., different noise rates or types among clients), where the client with severe noise would corrupt other clients and the FL framework ([Yang et al., 2022; Xu et al., 2022](#)). Therefore, the FL label noise should be tackled with global knowledge through the collaboration of the server and clients. Compared to the standard server aggregation ([McMahan et al., 2017](#)), a better solution is to leverage the learnable aggregation to improve the classifier with global knowledge for further client-side training. In this way, the generated classifier can better suit the entire FL framework under the joint server-client optimization.

To address the label noise problem in medical FL, we propose a FL framework with joint graph purification, named FedGP, to produce noise-robust diagnostic models through the server and clients collaboration. (1) For the client side, we first devise a noisy graph purification to reduce the interference of label noise on local training. This scheme can generate reliable pseudo labels for noisy samples by progressively expanding the purified graph with topological knowledge. With the client-side purified graph, we further propose a negative ensemble loss guided by the purified graph to provide robust complementary supervision of noisy samples. (2) For the server side, to combat label noise across decentralized clients, we propose a global centroid aggregation to produce a robust classifier with global knowledge, which can be optimized collaboratively in the FL framework. In this way, the proposed framework can effectively overcome the label noise problem in FL and provide a robust diagnostic model with superior performance.

The contributions of this work are summarized as follows:

- To handle label noise in medical FL, we propose a novel FedGP framework with joint graph purification under the server-client collaboration. To the best of our knowledge, this work represents the first effort to explore the label noise in medical FL.
- At the client side, we devise a noisy graph purification to generate reliable pseudo labels for noisy samples, by progressively expanding the purified graph from reliable clean samples. Moreover, a graph-guided negative ensemble (GNE) loss is proposed to complement the robust supervision of noisy samples with the purified graph.
- At the server side, we devise a global centroid aggregation to integrate the global knowledge across the FL framework to combat label noise under decentralized constraints. Through the collaborative optimization between server and clients, this strategy can produce a robust classifier with global knowledge against the FL label noise.
- Extensive experiments on endoscopic and pathological images prove the effectiveness, and our FedGP framework outperforms denoising and noisy FL state-of-the-arts under diverse FL noise settings by a large margin.

2. Related work

2.1. Federated learning

Federated Learning (FL) ([Kairouz et al., 2021](#)) provides a collaborative solution to optimize decentralized client models under the protection of private data. Particularly, FedAvg ([McMahan et al., 2017](#)) first proved that averaging local models of different clients is equivalent to updating the gradients in a centralized manner, with the assumption of independent and identically distributed (IID) clients. To alleviate the practical non-IID distribution between clients, FedProx ([Li et al., 2020b](#)) constrained the heterogeneous local updates close to the global model in the previous iteration, by introducing a parameter regularization term to the local training objective. To overcome the inter-client data heterogeneity, FedBN ([Li et al., 2021](#)) and SiloBN ([Andreux et al., 2020](#)) employed personalized normalization statistics for client models. FedProto ([Tan et al., 2022](#)) utilized category prototypes instead of local gradients to perform the server-client communication, and constrained the local prototypes to be sufficiently close to the global prototype aggregated on the server.

In computer-aided diagnosis, FL has attracted increasing attention with exploratory attempts at a variety of medical tasks ([Chen et al., 2021; Zhu et al., 2023](#)). For example, [Li et al. \(2019\)](#) implemented a practical FedAvg ([McMahan et al., 2017](#)) system with differential-privacy techniques on brain MRI segmentation. [Dou et al. \(2021\)](#) investigated the feasibility of detecting COVID-19 abnormalities from lung CT scans among multi-national hospitals. To utilize the knowledge of unlabeled data at the client side, [Yang et al. \(2021\)](#) extended the FL framework to semi-supervised CT segmentation by imposing consistency between pseudo labels and the predictions after input perturbations. [Chen et al. \(2022b\)](#) performed aggregation at frequency domain with adaptive knowledge transfer to address data heterogeneity and further aligned the server and client prototypes to overcome heterogeneous class imbalance in medical FL. Different from existing FL works in medical imaging that assume samples with absolutely clean labels, we propose a collaborative framework with joint graph purification to address the challenges of noisy labels in FL scenarios.

2.2. Learning with noisy labels

Label noise is a common problem in real-world datasets, since incorrect labels can result from the costly or difficult labeling process of annotating data ([Song et al., 2022](#)). Numerous methods have been devoted to learning with noisy labels, including heuristic strategies ([Gui et al., 2021; Li et al., 2020a](#)), noise transition matrix estimation ([Yao et al., 2020](#)), sample selection ([Han et al., 2018; Yu et al., 2019](#)), robust loss functions ([Wang et al., 2019; Zhou et al., 2021](#)) and negative learning ([Kim et al., 2019, 2021](#)). Particularly, some heuristic tricks treated a certain number of small-loss samples as clean ([Gui et al., 2021](#)), and found early stopping ([Li et al., 2020a; Xia et al., 2020](#)) can reduce the memorization of noise data. Noise transition matrix ([Yao et al., 2020](#)) can hardly be estimated in high noise rates or complicated instance-wise label noise, and sample selection works may discard noisy samples directly ([Han et al., 2018; Yu et al., 2019](#)), which ignore their supervision in high noise rate cases. [Wang et al. \(2019\)](#) devised

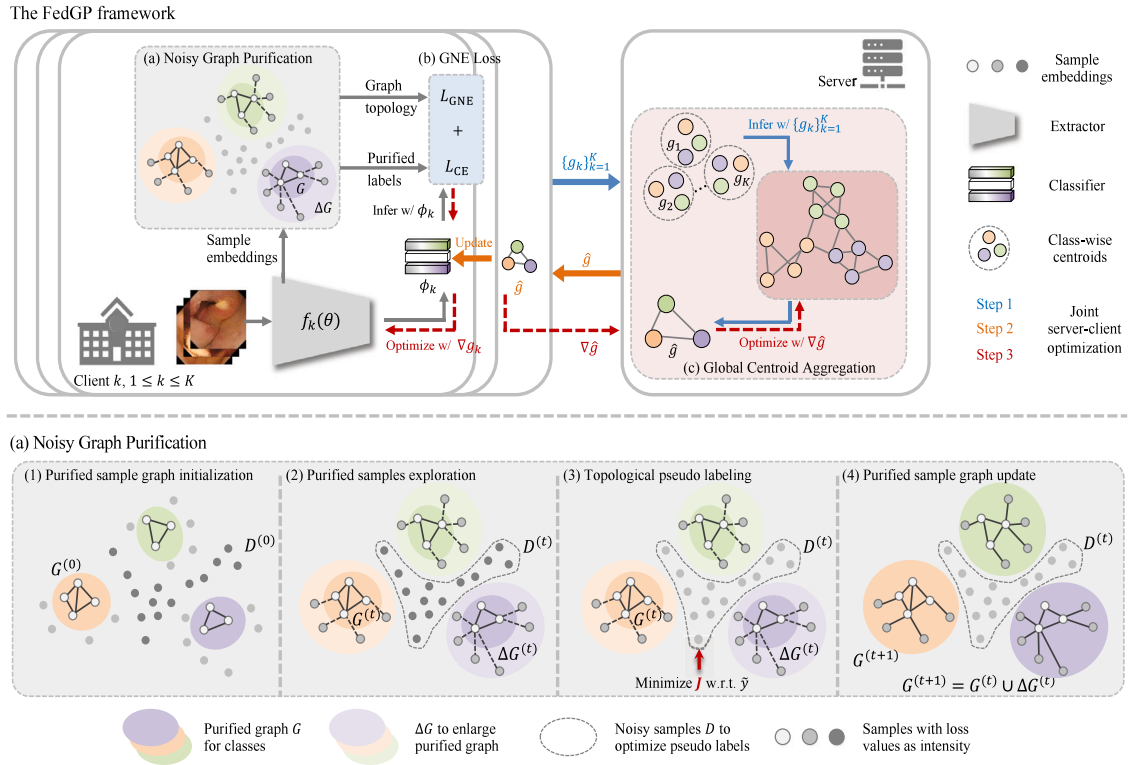


Fig. 1. The FedGP framework, including (a) Noisy Graph Purification, (b) Graph-guided Negative Ensemble (GNE) loss, and (c) Global Centroid Aggregation. Particularly, (a) Noisy Graph Purification at the client side with four iterative steps corrects noisy sample labels from reliable clean samples and progressively expands the purified sample graph, which exploits topological knowledge of sample relationship.

symmetric cross-entropy loss to ensure robust learning against label noise, and Zhou et al. (2021) proposed the sparse regularization for sharpening probabilities to further overcome the underfitting effect induced by the symmetric condition. Compared to positive learning that is directly affected by noisy labels, negative learning (Kim et al., 2019, 2021) utilized complementary labels to optimize the network training and reduced the risk of providing incorrect information. Yang et al. (2022) exchange the class centroids between the server and clients to address the label noise in FL case, and Xu et al. (2022) identify noisy samples at each client and leveraging all local data by gradually correcting the mistaken labels.

In the field of medical imaging, learning with noisy labels has recently become a cutting-edge and practical research topic, where medical labeling requires domain expertise and suffers from high intra-/inter-observer variability (Karimi et al., 2020). Among existing works, Xue et al. (2019) utilized an online uncertainty sample mining with sample re-weighting to overcome the impact of noisy data in an iterative manner. Liu et al. (2021) adopted the dual-network mutual learning to perform curriculum label correcting with estimated label probability. With the detection model to distinguish easy, hard and noisy samples, Zhu et al. (2021) devised a self-training strategy to correct noisy labels and enhance the informative hard samples for histopathology image classification. To the best of our knowledge, the proposed FedGP framework is the first work to address the label noise in medical FL scenarios.

3. Federated framework with joint graph purification

3.1. Overview of FedGP framework

Given medical FL with K clients, we denote the private data with noisy labels at k th client as $\{(x_i, y_i)\}_{i=1}^{N_k}$, where N_k is the sample size of k th client, and $y_i \in \mathbb{R}^C$ is the noisy label of sample x_i for C -class classification. The FedGP framework aims to produce a noise-resilient

global model $(\theta; \phi)$, which is parameterized with the feature extractor θ and the classifier ϕ . At the client side, to eliminate label noise during local training, the noisy graph purification (Section 3.2) progressively optimizes the pseudo labels $\{\tilde{y}_i\}_{i=1}^{N_k}$ of noisy samples and expands the purified sample graph. For the filtered noisy samples, the GNE loss (Section 3.3) further exploits the purified sample graph to complement robust supervision against label noise. At the server side, we devise the global centroid aggregation (Section 3.4) to integrate the client-side purified graph centroids with global knowledge and enhance the local classifier at each client. After each server-client communication, the server aggregates the feature extractors $\{\theta_k\}_{k=1}^K$ and the enhanced classifiers $\{\phi_k\}_{k=1}^K$ of local models through FedAvg (McMahan et al., 2017) for global model $(\theta; \phi)$, as $\theta = \frac{1}{K} \sum_{k=1}^K \theta_k$ and $\phi = \frac{1}{K} \sum_{k=1}^K \phi_k$. The proposed FedGP framework repeats these steps until communication reaches T rounds, as shown in Fig. 1.

3.2. Noisy graph purification in client

To combat label noise at the client side, we devise the noisy graph purification to progressively purify the rest noisy samples and correct these incorrect labels using the topological knowledge among the purified and noisy samples, as illustrated in Fig. 1(a). In FedGP framework, the noisy graph purification is performed starting from t_0 th FL round ($1 \leq t_0 \leq T$), which consists of four iterative operations as described below.

3.2.1. Purified sample graph initialization

According to the small loss criterion (Gui et al., 2021), we first select a few samples with the smallest loss values (e.g., $\gamma\%$ samples) as the reliable purified samples (i.e., scatter sample points with the lightest intensities in Fig. 1(a-1)), and initialize the purified sample graph $G^{(t_0)}$ with sample embeddings provided by the local feature extractor. Specifically, the purified sample graph is constructed among these purified samples belonging to the same category, as $G^{(t_0)} = \cup \{G_c^{(t_0)}\}_{c=1}^C$

in t_0 th round. For c th class, the subgraph $G_c^{(t_0)} = (V_c, E_c)$ regards sample embeddings of c th class serve as nodes V_c , and computes edges E_c between nodes based on cosine similarity, as $e_{ij} = v_i \cdot v_j / \|v_i\|_2 \|v_j\|_2$. Therefore, the initialized purified sample graph reflects the topological relationship of clean samples.

With the purified sample graph, we also calculate the class graph centroids $\{g_c\}_{c=1}^C$ by averaging the class-wise embeddings with edge weights. For example, the c th class graph centroid g_c is computed as follows:

$$g_c = \sum_{v_i \in G_c^{(t_0)}} \sum_{v_j \in \mathcal{N}_i^{(t_0)}} e_{ij} v_i, \quad (1)$$

where $\mathcal{N}_i^{(t_0)}$ represents the neighbor nodes of v_i in the purified sample graph $G^{(t_0)}$. As such, the class graph centroids $\{g_c\}_{c=1}^C$ can accurately represent the category information of the purified sample graph, and further provide a basis for pseudo labeling of noisy samples.

3.2.2. Purified samples exploration

Then, we explore the rest noisy samples to find reliable samples from two perspectives of output confidence and prediction consensus, as illustrated in Fig. 1(a-2). Compared with existing criteria (Liu et al., 2021), we select reliable samples as the purified ones with more rigorous conditions, including high-confidence output and consistent predictions. Specifically, for the t th round, the noisy samples $x \notin G^{(t)}$ with output confidence higher than τ can be regarded as clean, if the prediction by the local model is consistent with its optimized pseudo label (i.e., original noisy label in t_0 th round) and the prediction of class graph centroids. We denote these selected purified samples as $\Delta G^{(t)}$ in the t th round, which are further integrated into the purified sample graph $G^{(t)}$ at the end of this round (Fig. 1(a-4)). For the remaining noisy samples that do not meet the condition of $\Delta G^{(t)}$ (i.e., $x \notin G^{(t)} \cup \Delta G^{(t)}$), we denote them as the node set $D^{(t)}$ to be further optimized in Fig. 1(a-3). In this step, the rigorous conditions of both output confidence and prediction consensus can guarantee the quality of the purified sample graph $G^{(t)}$ at the client side.

3.2.3. Topological pseudo labeling

After that, we perform pseudo labeling for the noisy samples in $D^{(t)}$ with topological knowledge of the purified sample graph, as shown in Fig. 1(a-3). Specifically, we construct the connections W among the purified samples in $G^{(t)}$ and the noisy samples in $D^{(t)}$, which is implemented with nearest neighbors based on cosine similarity. With the topology priors among the samples in $G^{(t)}$ and $D^{(t)}$, we aim to improve the pseudo labels of samples in $D^{(t)}$ by optimizing the following objective:

$$\begin{aligned} \min_{\tilde{y}} J(\tilde{y}) := & (1 - \alpha_1 - \alpha_2) \sum_{i,j=1}^N W_{ij} \left\| \frac{\tilde{y}_i}{\sqrt{d_{ii}}} - \frac{\tilde{y}_j}{\sqrt{d_{jj}}} \right\|^2 \\ & + \alpha_1 (\|\tilde{Y} - Y_g\|_F^2) + \alpha_2 (\|\tilde{Y} - \tilde{Y}^{(t-1)}\|_F^2), \end{aligned} \quad (2)$$

where the first term encourages adjacent samples (i.e., similar node embeddings) to maintain consistent pseudo labels, the second term guides pseudo labels to keep consistent with the prediction Y_g of class graph centroids $\{g_c\}_{c=1}^C$, and the third term introduces the regularization with the previous pseudo labels $\tilde{Y}^{(t-1)}$. By optimizing Eq. (2), we produce high-quality pseudo labels with the topological knowledge of purified samples. In Fig. 1(a-2) of the next round, these optimized pseudo labels will decide whether to add the corresponding samples in $D^{(t)}$ to the purified sample graph.

3.2.4. Purified sample graph update

At the end of each round, we update the purified sample graph $G^{(t)}$ with newly-added reliable samples $\Delta G^{(t)}$ and re-compute the class graph

centroids $\{g_c^{(t)}\}_{c=1}^C$ using the improved graph topology $G^{(t+1)}$, as follows:

$$\begin{aligned} G_c^{(t+1)} &= G_c^{(t)} \cup \Delta G_c^{(t)}, \\ g_c^{(t+1)} &= \sum_{v_i \in G_c^{(t+1)}} \sum_{v_j \in \mathcal{N}_i^{(t+1)}} e_{ij} v_i. \end{aligned} \quad (3)$$

By iteratively performing these aforementioned operations, our method progressively expands the purified sample graph and generates reliable pseudo labels for noisy samples, until the graph structure and labels remain stable during the FL training.

Different from the existing label propagation work (Wu et al., 2021) that first builds a graph with all samples and then prunes unreliable nodes and edges, our noisy graph purification constructs the purified graph with reliable core samples and gradually adds more purified samples with multiple criteria. Therefore, our strategy makes the propagation of pseudo labels less affected by noisy samples, thus benefiting the FL framework even more.

3.3. Graph-guided negative ensemble loss in client

In terms of noisy samples outside the purified sample graph (i.e., $x \notin G^{(t)}$), the proposed noisy graph purification in Section 3.2 can effectively exclude the interference of these mistaken labels on the local training of FL. But this strategy would lose the supervision of massive local data with noisy labels, and thus cannot fully utilize the data knowledge of the entire FL framework to combat label noise. To address this dilemma, we propose a graph-guided negative ensemble (GNE) loss to generate complementary labels for noisy samples and suppress the corresponding predictions. In this way, the proposed FedGP framework can leverage the supervision of both the purified sample graph and the outside noisy samples, and the overall loss function is summarized as follows:

$$L = \sum_{x \in G} L_{CE} + \lambda_{GNE} \sum_{x \notin G} L_{GNE}, \quad (4)$$

where the cross-entropy loss L_{CE} supervises reliable samples $x \in G$, and our L_{GNE} with the weight λ_{GNE} provides auxiliary supervision with the remaining noisy samples $x \notin G$.

To mitigate this risk with robust and extensive supervision against label noise, the proposed GNE loss exploits topological knowledge of purified sample graph at the client side. To enrich the insufficient supervision information provided by existing negative learning techniques (Kim et al., 2019, 2021), we leverage the well-established purified graph with neighborhood knowledge to promote the negative supervision. Specifically, we generate multiple complementary labels $\tilde{Y}_i$ for i th noisy sample as the ensemble of negative learning, as follows:

$$L_{GNE}(f, \tilde{y}_c) = - \sum_{c=1}^{C_{comp}} \tilde{y}_c \log(1 - p_c), \quad (5)$$

where multiple complementary labels $\tilde{Y}_i = \{\tilde{y}_c\}_{c=1}^{C_{comp}}$ are selected by the graph knowledge, and can provide richer supervision information for each noisy sample. Note that existing negative learning strategies (Kim et al., 2019, 2021) randomly selected a single complementary label for each noisy sample, which poses the risk of incorrect and limited information. In contrast, our GNE loss utilizes the purified graph topology to select multiple reliable labels based on the neighborhood consensus, which can reduce the risk of mistakenly selecting the true label. In this way, the proposed GNE loss can provide more reliable and sufficient supervision knowledge for the joint optimization of noisy FL.

3.4. Global centroid aggregation in server

Although tailored denoising techniques can eliminate the impact of label noise on local training, these techniques are limited to private data at the client side and cannot fully leverage the knowledge

of the entire FL framework. In addition, the server aggregation in FL may introduce noisy information from other clients and cannot handle the heterogeneous label noise, thereby leading to unsatisfying performance. To gather global knowledge across clients against FL label noise, we propose a global centroid aggregation at the server side to produce classifier parameters for noise-robust models with superior performance.

3.4.1. Global centroids generation

With the noisy graph purification and the GNE loss to combat the label noise at the client side, the graph centroids $\{\mathbf{g}_k\}_{k=1}^K$ can reflect the class-wise characteristics of clean samples at different clients, where $\mathbf{g} = [g_1, g_2, \dots, g_C]$ is the centroids at each client. Therefore, the proposed global centroid aggregation gathers these graph centroids from different clients to exploit the noise-robust knowledge, which can be jointly optimized through the server and clients collaboration. With the graph centroids $\{\mathbf{g}_k\}_{k=1}^K$ of different clients as input nodes, the global centroid aggregation can exploit the relationship among these centroids from different clients, and generate the output centroid $\hat{\mathbf{g}}$ integrated with the global knowledge of the FL framework, as follows:

$$\hat{\mathbf{g}} = \mathcal{A}_{\text{GCent}}(\{\mathbf{g}_k\}_{k=1}^K). \quad (6)$$

In our implementation, the global centroid aggregation $\mathcal{A}_{\text{GCent}}$ consists of two graph convolutional layers with non-linear activation function (e.g., ReLU), where the adjacency matrix is calculated with cosine similarity between every two centroids. In this way, the global centroid $\hat{\mathbf{g}}$ can be transmitted to each client, and provides local classifiers with the global knowledge against FL label noise.

3.4.2. Joint server-client optimization

To guarantee the global centroid beneficial for the local models at clients, we devise a joint optimization strategy for the global centroid aggregation through the collaboration between the server and clients. As illustrated in Fig. 1(c), the joint optimization of the aggregation is performed in three steps. First, we compute the graph centroid with the purified samples in Eq. (1) at the client side, and the graph centroids $\{\mathbf{g}_k\}_{k=1}^K$ of different clients are delivered to the server, where the global centroid $\hat{\mathbf{g}}$ is generated by the proposed global centroid aggregation in Eq. (6).

Then, the server distributes the global centroid $\hat{\mathbf{g}}$ to each client, and the classifier parameters of the local model are mixed with two components (i.e., local classifier parameters ϕ_k and global centroid $\hat{\mathbf{g}}$) at the end of communication, as follows:

$$\phi_k^{(t+1)} := (1 - \beta)\phi_k^{(t)} + \beta\hat{\mathbf{g}}^{(t)}, \quad (7)$$

where $\phi_k^{(t)}$ is the previous local parameters, $\hat{\mathbf{g}}^{(t)}$ is the global parameters generated by the global centroid aggregation, and β is the hyper-parameter to control the proportion of global knowledge. In this way, the local model can integrate the global knowledge from the server to perform the local training over the private data. For the targeted global model, we average the local classifiers $\{\phi_k^{(t+1)}\}_{k=1}^K$ to generate the global classifier $\phi^{(t+1)}$, as $\phi^{(t+1)} = \frac{1}{K} \sum_{k=1}^K \phi_k^{(t+1)}$.

Finally, with the local classifier updated in Eq. (7), the global centroid can receive the gradient $\beta \nabla \mathbf{g}_k^{(t+1)}$ during the back propagation of local training. As such, the FedGP framework can optimize the global centroid aggregation with $\nabla \hat{\mathbf{g}} = \frac{1}{K} \sum_{k=1}^K \nabla \mathbf{g}_k$ from clients, and facilitate the server-clients collaboration in an iterative manner. Note that the average gradients $\nabla \hat{\mathbf{g}}$ uploaded to the server do not involve privacy, because the irreversible averaging operation erases the private information of each sample (Ma et al., 2022; Shang et al., 2022).

Algorithm 1 The pipeline of FedGP framework.

Input : The FL dataset with K clients;
The network with extractor θ and classifier ϕ ;
Communication iteration T and t_0 ;

Output: The trained diagnostic model with θ and ϕ .

Server executes:

- 1: Initialize the model parameters θ and ϕ for all clients;
- 2: **for** each round t in T **do**
- 3: **for** each client k in **parallel do**
- 4: $\theta_k, \phi_k, \mathbf{g}_k \leftarrow \text{LocalUpdate}(t, k, \theta, \phi, \hat{\mathbf{g}})$;
- 5: **end for**
- 6: Update θ with $\{\theta_k\}_{k=1}^K$ and ϕ with $\{\phi_k\}_{k=1}^K$ via FedAvg;
- 7: Update global centroid $\hat{\mathbf{g}}$ with $\{\mathbf{g}_k\}_{k=1}^K$ via global centroid aggregation $\mathcal{A}_{\text{GCent}}$ in Eq. (6);
- 8: **end for**
- 9: **return** well-trained θ and ϕ for FedGP

LocalUpdate($t, k, \theta, \phi, \hat{\mathbf{g}}$):

- 1: Update θ_k with θ and ϕ_k with ϕ ;
- 2: **if** $t < t_0$ **then**
- 3: Optimize θ_k and ϕ_k with L_{CE} to warm-up FedGP;
- 4: **else**
- 5: Perform noisy graph purification;
- 6: Optimize θ_k and ϕ_k with L_{CE} and L_{GNE} in Eq. (5);
- 7: **end if**
- 8: Update ϕ_k with $\hat{\mathbf{g}}$ in Eq. (7) and class centroids $\mathbf{g}_k$;
- 9: **return** θ_k, ϕ_k and $\mathbf{g}_k$

3.5. Optimization pipeline

We summarize the proposed FedGP framework in Algorithm 1. On the basis of FedAvg (McMahan et al., 2017), we first merely perform the global centroid aggregation (Section 3.4) at server during the first t_0 rounds, which warms up the FedGP framework and integrates client models with global knowledge. At the client side since t_0 th round, we also perform the noisy graph purification (Section 3.2) to correct the labels of noisy samples, and compute the GNE loss (Section 3.3) over the remaining noisy samples to provide auxiliary supervision. During each round after the warm-up, the FedGP framework progressively enlarges the purified sample graph at the client side, and generates the global centroids at the server side to promote model classifiers with joint server-client optimization. In this way, the FedGP framework overcomes the FL label noise in a collaborative manner, and finally produces a global model with superior performance.

4. Experiments

4.1. Dataset and implementation

4.1.1. Endoscopy FL dataset with synthetic label noise

To perform evaluation under FL label noise, we collect endoscopy samples of ten largest categories from Kvasir-Capsule dataset (Smedsrud et al., 2021) to construct the endoscopy FL dataset, which contains 3000 normal, 3000 ileocecal valve, 2906 reduced mucosal, 1529 pylorus, 866 angiectasia, 854 ulcer, 776 foreign body, 592 lymphangiectasia, 506 erosion and 446 blood samples. We split these samples into training, validation and test sets as 7:1:2, and then equally divide the training set into four clients to formulate the FL scenario. To investigate the impact of label noise, we employ the symmetric and pairflip label noise (Han et al., 2018) with diverse settings on training set, which is widely-used in both natural and medical imaging communities (Han et al., 2018; Zhu et al., 2021). Therefore, this dataset with synthetic label noise provides the feasibility to comprehensively compare state-of-the-art methods and conduct ablation analysis under diverse FL label noise settings.

Table 1Comparison on endoscopy FL dataset under homogeneous label noise. Best and second best results are **highlighted** and underlined.

Method	Symm. Noise						Pairflip noise					
	10%			20%			10%			20%		
	F1	Recall	Prec.	F1	Recall	Prec.	F1	Recall	Prec.	F1	Recall	Prec.
FedAvg (McMahan et al.)	85.20	84.61	90.28	79.11	79.56	78.95	81.59	82.51	89.45	74.36	75.03	83.03
FedProx (Li et al.)	85.18	84.42	90.31	81.30	80.98	88.74	82.02	82.71	89.94	75.15	75.50	83.33
SymmCE (Wang et al.)	90.50	90.86	90.18	81.12	80.88	88.65	83.09	82.17	78.30	73.04	74.28	73.06
LNL-SR (Zhou et al.)	89.12	88.38	90.04	84.56	<u>83.61</u>	85.69	<u>83.76</u>	<u>84.86</u>	82.79	<u>78.32</u>	<u>79.78</u>	77.00
FedRobust (Yang et al.)	87.12	85.96	88.99	80.74	79.82	84.00	81.20	81.56	88.84	71.89	72.23	81.99
FedCorr (Xu et al.)	87.64	85.51	<u>91.75</u>	<u>84.68</u>	82.78	<u>89.60</u>	82.35	82.32	<u>90.69</u>	74.12	74.10	74.95
FedGP (Ours)	91.33	<u>90.45</u>	92.51	89.81	89.41	90.41	89.19	88.04	90.75	86.30	85.14	88.20

4.1.2. Histopathology FL dataset with real-world label noise

The histopathology image dataset with real-world label noise is collected from Chaoyang hospital (Zhu et al., 2021), which includes 1816 normal, 1163 serrated, 2244 adenocarcinoma, 937 adenoma samples of colon slides. The dataset selects 705 normal, 321 serrated, 840 adenocarcinoma, 273 adenoma samples with the consensus of 3 pathologists as test set, and treats the rest samples as training set, by randomly selecting one opinion from three pathologists as the noisy label. With the label noise annotated by pathologists, we further construct the FL dataset by equally dividing the training set into three clients. As such, this histopathology FL dataset can validate the effectiveness of FL algorithms under real-world label noise.

4.1.3. Implementation details

We implement our FedGP framework and state-of-the-art FL approaches on NVIDIA RTX 2080ti GPU using PyTorch (Paszke et al., 2019), and adopt VGG-11 (Simonyan and Zisserman, 2015) as diagnostic model for FL evaluation. In the comparison of both datasets, the networks in the proposed FedGP framework and state-of-the-art methods are optimized with SGD for $T = 100$ iterations with the batch size of 128 and the weight decay of 1×10^{-6} . The initial learning rate is set as 1×10^{-2} , and learning rate is divided by 10 at 70% and 90% progress of local training to ensure convergence. In the global centroid aggregation, we set $\beta = 0.5$ to control the classifier mixture of local parameters and global centroids, and we also adopt the global centroid aggregation to warm up the FedGP framework at the first $t_0 = 20$ rounds. In the noisy graph purification, the initialized purified samples γ is 20%, the confidence threshold τ is set as 0.6, and $\alpha_1 = \alpha_2 = 0.5$. In the GNE loss, λ_{GNE} is set as 0.1, and we set the node neighborhood as 20 nodes and $C_{\text{comp}} = C - 2$ for reliable complementary labels, i.e., 8 and 2 for endoscopy and histopathology datasets, respectively. The input images are unified with the resolution of 128×128 in both FL datasets, and the local training is augmented with random flip and rotation of input images to avoid overfitting. All methods in the experiments use the same random seed to ensure fair comparisons and reproducible performance.

4.1.4. Evaluation metrics

To evaluate the performance with FL label noise, we adopt the F1 score (F1), recall and precision to compare various FL methods comprehensively. Specifically, recall and precision evaluate the model predictions in terms of completeness and quality, respectively, while F1 demonstrates the general evaluation using the harmonic average of these two aspects. For the multi-category diagnosis, we first compute the metric for each class individually, and then calculate the macro-average to consider each category equally (Zhu et al., 2021). The higher F1, recall and precision values reveal superior diagnostic performance on both endoscopy and histopathology FL datasets.

4.2. Comparison under homogeneous FL label noise

On the endoscopy FL dataset, we compare our FedGP framework with state-of-the-art works from diverse perspectives. In particular,

Table 2Comparison on endoscopy FL dataset under extreme symmetric label noise. Best and second best results are **highlighted** and underlined.

Method	40%			60%		
	F1	Recall	Prec.	F1	Recall	Prec.
FedAvg	65.38	66.90	70.07	45.11	46.77	47.93
FedProx	66.78	68.26	70.35	46.03	48.47	48.76
SymmCE	70.53	70.92	70.97	43.66	47.19	42.56
LNL-SR	71.96	72.19	74.31	56.72	57.19	59.44
FedRobust	62.33	63.66	66.60	50.80	50.02	55.75
FedCorr	62.74	63.04	64.04	52.75	51.76	55.76
FedGP	82.45	81.78	84.37	66.78	68.18	66.88

FedAvg (McMahan et al., 2017) and FedProx (Li et al., 2020b) serve as the baseline of FL frameworks under FL label noise. Also, the prevalent label denoising works, i.e., SymmCE (Wang et al., 2019) and LNL-SR (Zhou et al., 2021), are implemented on the basis of FedAvg (McMahan et al., 2017) for the FL scenario, which are employed to combat the label noise in local training at each client. Moreover, we further compare the advanced FedRobust (Yang et al., 2022) and FedCorr (Xu et al., 2022) targeted for FL label noise. In this section, we conduct comprehensive comparison with different homogeneous label noise cases in Section 4.2.1 and the extreme label noise cases in Section 4.2.2.

4.2.1. Homogeneous FL label noise

To validate the effectiveness of the FedGP framework at different label noise scenarios, we perform the comparison under symmetric and pairflip label noise with noise rate of 10% and 20%, as illustrated in Table 1. In general, these experimental results reveal consistent performance trends in comparisons with symmetric label noise. Under 20% symmetric label noise, FedAvg (McMahan et al., 2017) serves as the FL baseline with 79.11% F1. On this basis, SymmCE (Wang et al., 2019) (F1: 81.12%) and LNL-SR (Zhou et al., 2021) (F1: 84.56%) alleviate the impact of FL label noise on local training to improve the performance. Furthermore, FedCorr (Xu et al., 2022) conducts local training using identified clean samples, and corrects noisy labels of local samples with client collaboration, which achieves the second-best F1 of 84.86%. With the tailored designs in both client and server sides, our FedGP framework performs the robust local training and effective FL aggregation, thereby leading to the best performance with F1 of 89.91%.

Compared to symmetric label noise, the pairflip label noise is more difficult, where the errors are concentrated in a certain class. Under pairflip label noise, LNL-SR (Zhou et al., 2021) reveals superior performance over other state-of-the-art works, and achieves the second-best F1 of 83.76% and 78.32% for 10% and 20% cases, respectively. This confirms that high-quality local training on the client side is effective in mitigating FL homogeneous label noise. In contrast, our FedGP framework achieves the best performance in both noise rates, with F1 of 89.19% and 86.30% in 10% and 20% label noise respectively. Especially, the FedGP framework significantly improves the performance with an increase of 7.98% F1, 5.36% recall and 4.87% precision

Table 3Comparison on endoscopy FL dataset under label noise with heterogeneous rates. Best and second best results are **highlighted** and underlined.

Method	Symm. Noise						Pairflip noise					
	10%-10%-20%-20%			10%-20%-20%-40%			10%-10%-20%-20%			10%-20%-20%-40%		
	F1	Recall	Prec.	F1	Recall	Prec.	F1	Recall	Prec.	F1	Recall	Prec.
FedAvg (McMahan et al.)	83.33	82.86	90.05	81.51	81.54	86.01	80.25	81.20	79.44	76.79	<u>77.75</u>	76.67
FedProx (Li et al.)	83.98	83.49	90.30	81.74	81.59	86.51	80.13	81.02	79.41	75.80	76.86	76.15
SymmCE (Wang et al.)	87.07	87.10	87.38	75.04	77.94	82.02	77.42	80.26	75.06	63.65	67.07	68.84
LNL-SR (Zhou et al.)	<u>87.16</u>	<u>86.16</u>	88.44	<u>82.32</u>	<u>82.29</u>	82.56	<u>85.45</u>	<u>84.33</u>	87.10	<u>78.15</u>	76.62	80.46
FedRobust (Yang et al.)	85.05	83.77	87.11	80.57	80.44	83.15	79.41	79.13	87.30	70.37	70.94	81.43
FedCorr (Xu et al.)	86.18	84.97	89.15	81.81	80.59	<u>87.27</u>	82.15	81.05	<u>89.58</u>	74.16	74.29	<u>83.34</u>
FedGP (Ours)	90.40	89.34	91.84	89.01	88.10	90.93	88.59	87.48	90.11	85.68	84.72	86.98

Table 4Comparison on endoscopy FL dataset under label noise with heterogeneous types. Best and second best results are **highlighted** and underlined.

Method	S10%-S10%-P10%-P10%			S10%-S20%-P10%-P20%			S20%-S20%-P20%-P20%		
	F1	Recall	Prec.	F1	Recall	Prec.	F1	Recall	Prec.
FedAvg (McMahan et al., 2017)	82.06	82.72	90.91	80.45	81.36	79.65	77.21	77.75	77.34
FedProx (Li et al., 2020b)	82.14	82.98	90.80	80.22	80.89	79.63	77.45	77.83	77.83
SymmCE (Wang et al., 2019)	83.25	83.95	82.70	80.14	80.44	80.43	79.88	80.88	79.37
LNL-SR (Zhou et al., 2021)	83.66	82.25	<u>91.42</u>	81.32	80.65	<u>89.03</u>	80.26	79.60	<u>87.85</u>
FedRobust (Yang et al., 2022)	84.43	83.32	90.52	79.99	80.34	86.67	79.53	79.19	84.15
FedCorr (Xu et al., 2022)	84.38	<u>85.47</u>	83.46	<u>83.18</u>	<u>83.33</u>	83.23	<u>81.80</u>	<u>82.13</u>	81.76
FedGP (Ours)	91.08	90.11	92.38	90.07	88.98	91.59	86.74	85.86	88.37

over the second-best method in 20% noise rate. These comparisons of symmetric and pairflip label noise verify the consistent and significant superiority of our FedGP framework in different homogeneous FL label noise scenarios.

4.2.2. Extreme FL label noise

We further validate the effectiveness of the FedGP framework under extreme FL label noise, including 40% and 60% symmetric label noise. As illustrated in Table 2, our FedGP framework achieves the best performance in both extreme label noise cases, and outperforms the second-best LNL-SR (Zhou et al., 2021) with a F1 margin of 10.49% and 10.06% in 40% and 60% symmetric label noise, respectively. It is worth noting that our FedGP framework still achieves ideal diagnostic performance with 82.45% F1 at 40% symmetric label noise, even higher than most state-of-the-art methods (McMahan et al., 2017; Li et al., 2020b; Wang et al., 2019; Yang et al., 2022) at 20% symmetric label noise in Table 1. From these comparisons, we confirm that the proposed FedGP framework is resistant to extreme FL label noise with higher rates over state-of-the-art denoising works.

4.3. Comparison under heterogeneous FL label noise

In addition to the homogeneous label noise comparison, we further perform comprehensive evaluation under heterogeneous FL label noise on the endoscopy FL dataset. In particular, we setup the medical FL scenarios with heterogeneous label noise from two perspectives, including heterogeneous noise rates in Section 4.3.1 and heterogeneous noise types in Section 4.3.2.

4.3.1. Heterogeneous FL label noise rates

To evaluate the performance under heterogeneous noise rates, we perform the comparison in both symmetric and pairflip label noise with noise rates as 10%-10%-20%-20% and 10%-10%-20%-40% for four clients. Particularly, under the symmetric label noise with 10%-10%-20%-40% rates, our FedGP framework achieves the best performance (F1: 89.01%) with the balanced recall (88.10%) and precision (90.93%). Compared with the second-best LNL-SR (Zhou et al., 2021), our FedGP framework obtains a performance gap with 6.69% in F1, 5.81% in recall and 8.37% in precision, which reveals a more significant advantage in the homogeneous comparison (Table 1). Moreover, our FedGP framework also achieves consistent performance advantages

with other heterogeneous noise rates, including the 10%-10%-20%-20% noise rates and the pairflip noise as shown in Table 3. Therefore, these comparisons confirm the superiority of our FedGP framework in the medical FL scenarios with heterogeneous noise rates.

4.3.2. Heterogeneous FL label noise types

To investigate the impact of heterogeneous noise types, we further implement our FedGP framework and state-of-the-art methods in two symmetric noise clients and two pairflip noise clients, where S and P in Table 4 represent symmetric and pairflip label noise, respectively. In particular, FedCorr (Xu et al., 2022) outperforms other existing works in three cases with heterogeneous noise types, which verifies that client collaboration is important to combat the FL label noise with heterogeneous types. In contrast, our FedGP framework achieves the best performance with the F1 of 91.08%, 90.07% and 86.74% in three cases with heterogeneous noise types, and outperforms FedCorr (Xu et al., 2022) with the F1 gap of 6.70%, 6.89% and 4.94%, respectively. These comparisons confirm that the proposed FedGP framework can utilize the global knowledge of clients to overcome the heterogeneous FL label noise, including heterogeneous noise rates and types.

4.4. Comparison under real-world FL label noise

To validate the effectiveness of the proposed FedGP framework under real-world FL label noise, we perform the comparison on the histopathology FL dataset (Zhu et al., 2021). As shown in Table 6, FedAvg (McMahan et al., 2017) with F1 of 71.36% serves as the FL baseline for the real-world label noise setup, and FedProx (Li et al., 2020b) (F1: 72.37%) slightly alleviates the impact of label noise through the parameter regularization on local training. On this basis, existing denoising methods (Wang et al., 2019; Zhou et al., 2021; Yang et al., 2022; Xu et al., 2022) effectively combat the real-world label noise with tailored designs, especially FedCorr (Xu et al., 2022) with the F1 of 75.74%. The proposed FedGP framework achieves the best performance with F1 of 76.84%, and achieves the performance improvement of 5.54% in recall and 5.89% in precision compared with the FedAvg baseline (McMahan et al., 2017). From these comparisons, we comprehensively confirm the superiority of our FedGP framework under both synthetic and real-world FL label noise.

Table 5
Ablation study on endoscopy FL dataset under 20% symmetric/pairflip label noise.

	FedAvg	$\mathcal{A}_{\text{GCent}}$	Noisy graph purif.	L_{GNE}	Symm. Noise			Pairflip noise		
					F1	Recall	Prec.	F1	Recall	Prec.
1	✓				79.11	79.56	78.95	74.36	75.03	83.03
2	✓	✓			82.58	81.21	84.65	78.71	78.38	79.45
3	✓	✓	✓		88.42	87.89	89.10	84.79	84.41	85.55
4	✓	✓	✓	✓	89.81	89.41	90.41	86.30	85.14	88.20

Table 6
Comparison on histopathology FL dataset under real-world label noise. Best and second best results are **highlighted** and underlined.

Method	F1	Recall	Prec.
FedAvg	71.36	70.25	73.14
FedProx	72.37	71.76	73.18
SymmCE	74.24	73.47	76.50
LNL-SR	74.90	73.86	76.51
FedRobust	74.80	73.74	76.35
FedCorr	<u>75.74</u>	<u>74.75</u>	<u>77.13</u>
FedGP	76.84	75.79	79.03

Table 7
Ablation study on histopathology FL dataset under real-world label noise.

	FedAvg	$\mathcal{A}_{\text{GCent}}$	NGP	L_{GNE}	F1	Recall	Prec.
1	✓				71.36	70.25	73.14
2	✓	✓			73.53	72.95	74.42
3	✓	✓	✓		75.27	75.41	75.60
4	✓	✓	✓	✓	76.84	75.79	79.03

4.5. Ablation study

4.5.1. Effects of tailored components

To quantitatively analyze the contribution of the proposed components (i.e., global centroid aggregation $\mathcal{A}_{\text{GCent}}$, noisy graph purification, GNE loss L_{GNE}) to combat the FL label noise, we perform the ablation study of the proposed FedGP framework under both 20% symmetric and pairflip label noise, as shown in Table 5. Compared with the FedAvg baseline (Line 1), the global centroid aggregation $\mathcal{A}_{\text{GCent}}$ (Line 2) achieves the F1 of 82.58% and 78.71% in symmetric and pairflip label noise respectively, which results in a F1 increase of 3.47% and 4.35%. This increase confirms that the joint optimized class centroids can promote the classifier with global knowledge. Moreover, the noisy graph purification (Line 3) with the F1 of 88.42% and 84.79%, significantly improves the performance with a F1 increase of 5.84% and 6.08% in symmetric and pairflip label noise respectively. This advantage validates the superiority of the topology-guided purification to combat the label noise at the client side. Finally, the GNE loss L_{GNE} (Line 4) further complements the supervision of noisy samples to facilitate the local training, which brings an additional F1 improvement of 1.39% and 1.51%. Furthermore, to validate the effectiveness under real-world label noise, we also analyze these tailor components on the histopathology FL dataset. The results in Table 7 elaborate the effectiveness under real-world label noise with consistent improvements, where NGP in Table 7 stands for the noisy graph purification. These ablation experiments validate that the tailor-made modules of FedGP framework can comprehensively facilitate the medical FL diagnosis against label noise.

4.5.2. Analysis of noisy graph purification

To further investigate the noisy graph purification, we analyze the training dynamics of the purified sample graph on endoscopy FL dataset with 20% symmetric label noise. As illustrated in Fig. 2(a), the purified sample graph starts with the predefined $\gamma = 20\%$ reliable samples, and rapidly purifies the remaining noisy samples from 20th to 30th round. The purified sample graph converges at 50th round with almost all samples added. This confirms that the noisy graph purification

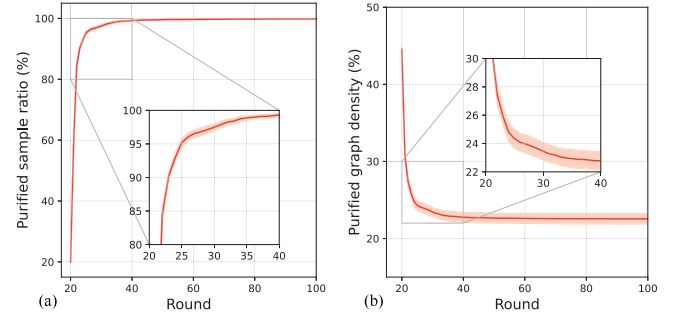


Fig. 2. The analysis of noisy graph purification, including (a) the ratio of purified samples, and (b) the density of purified sample graph. The curve reflects the average of four clients in endoscopy FL dataset, and the shaded area represents the standard deviation between four clients.

Table 8
Performance and communication analysis of local centroid communication on endoscopy FL dataset under 20% symmetric label noise.

Interval	Performance		Communication	
	F1 (%)	ΔF1 (%)	Amount (MB)	Overhead (%)
FedAvg	79.11	–	1.417	–
8	84.28	5.16	1.423	0.43
4	87.72	8.61	1.429	0.86
2	89.23	10.12	1.442	1.72
1	89.81	10.70	1.466	3.44

can quickly provide the topological knowledge of local dataset for FL framework. Moreover, with the enlargement of the purified sample graph, the density decreases from 44% of the initialized graph and quickly converges to 22% after 40th round, which indicates the sparse purified sample graph remains efficient during the FL. In this way, the noisy graph purification can efficiently support robust FL against label noise.

4.5.3. Analysis of local centroid communication

Considering the FedGP framework induces the local centroid communication, we further analyze the trade-off of performance and communication overhead on endoscopy FL dataset with 20% symmetric label noise. Specifically, we implement the FedGP framework with different communication intervals of local centroids between server and clients, where a larger interval can reduce more communication overhead induced by local centroids. As illustrated in Table 8, the default FedGP (interval = 1) significantly improves the performance with a 10.70% F1 increase at the cost of merely a 3.44% communication overhead, compared to the FedAvg (McMahan et al., 2017) baseline. In practice, this communication overhead is negligible for medical FL systems. Furthermore, the FedGP framework can reduce the communication to meet the resource requirement with specific intervals, e.g., FedGP achieves the F1 of 89.23% and 87.72% with 1.72% and 0.86% communication overhead, respectively. Therefore, the proposed FedGP framework only involves a slight communication overhead, and achieves an efficient trade-off of performance and communication for medical FL under label noise.

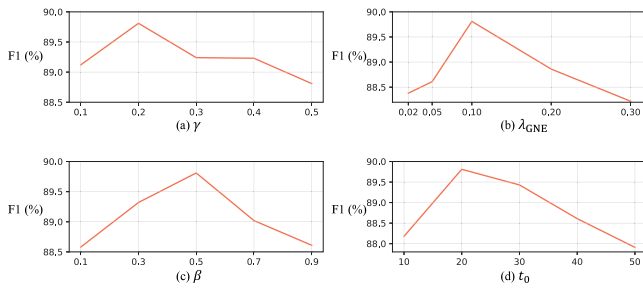


Fig. 3. The analysis of hyper-parameters, including (a) the initialized sample ratio γ in noisy graph purification, (b) the loss weight λ_{GNE} of the GNE loss, (c) the proportion β of global centroid in the global centroid aggregation, and (d) the warm-up round t_0 of the global centroid aggregation.

Table 9

Comparison of different local epochs in FL on endoscopy FL dataset under 20% symmetric label noise.

Local epoch	FedAvg	FedCorr	FedGP
Default = 1	81.48	84.68	89.81
2	81.47	84.33	89.43
3	81.22	82.99	88.16
4	81.03	82.19	86.17
5	80.75	82.38	85.95
10	80.06	81.30	83.70

4.5.4. Analysis of hyper-parameters

We investigate the impact of hyper-parameters of the proposed FedGP framework in this section. Specifically, we perform grid search on the most critical hyper-parameter in each module, including the initialized sample ratio γ in noisy graph purification, the loss weight λ_{GNE} of the GNE loss, and the proportion β of global centroid in the global centroid aggregation, as well as the warm-up round t_0 to balance three modules. The results in Fig. 3 confirm that we select rational values for the FedGP framework, and FedGP consistently outperforms state-of-the-arts under different values. It is noteworthy that the warm-up round t_0 has a greater impact on performance among these hyper-parameters in the FedGP framework. In addition to the default FL setting, we further study the performance gap with different local epochs in FL. In Table 9, our FedGP framework significantly performs better than FedAvg (McMahan et al., 2017) and the second-best FedCorr (Xu et al., 2022) with local epoch from the default 1 to 10. From these comparisons, we prove that our FedGP framework is robust to achieve superior performance with different methodological and FL hyper-parameters.

4.5.5. Visualization comparison

As illustrated in Fig. 4, we further present the t-SNE (Maaten and Hinton, 2008) visualization of output space to quantitatively analyze the advantage of FedGP over FedAvg (McMahan et al., 2017) and FedCorr (Xu et al., 2022) on endoscopy FL dataset with 20% symmetric label noise. Compared with FedGP in Fig. 4(d), FedAvg in Fig. 4(a) fails to distinguish the 7th (lymphangiectasia) and 8th (erosion) minority categories, and FedCorr in Fig. 4(b) cannot handle the intra-class variance of 1st category (ileocecal valve) and results in unsatisfactory clusters. Moreover, our FedGP can produce high-quality t-SNE distribution at only 50th round in Fig. 4(c), and further refines the diagnostic model with the best results in Fig. 4(d). These comparisons quantitatively prove the superiority of our FedGP framework in FL label noise.

5. Conclusion

In this work, we systematically analyze the label noise problem in medical FL scenarios, which severely degrades the performance of

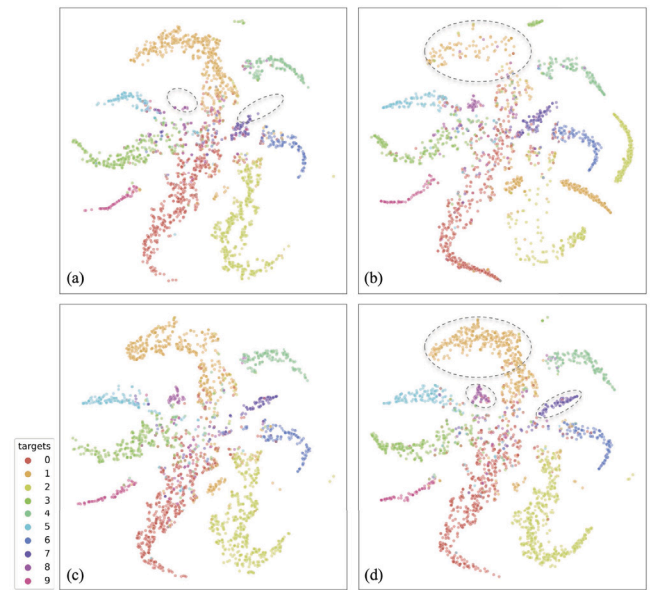


Fig. 4. The t-SNE comparison of (a) FedAvg, (b) FedCorr, (c) FedGP at 50th round, and (d) FedGP. Particularly, (a), (b) and (d) are well-trained models.

existing FL approaches. To combat the label noise, we propose a novel FL framework with joint Graph Purification (FedGP) to produce noise-robust diagnostic models under server and clients collaboration. At the client side, we devise a noisy graph purification to generate reliable pseudo labels with topological knowledge by progressively expanding the purified sample graph. Then, we further propose a graph-guided negative ensemble loss to exploit the topology of the client-side purified graph with robust complementary supervision against label noise. At the server side, we propose a global centroid aggregation to produce the robust classifier with global knowledge, which can address the data silos in FL with collaborative optimization. Extensive experiments on endoscopy and histopathology FL datasets demonstrate the remarkable advantages of our FedGP framework over state-of-the-art approaches.

CRedit authorship contribution statement

Zhen Chen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – review & editing. **Wuyang Li:** Conceptualization, Methodology, Writing – review & editing. **Xiaohan Xing:** Conceptualization, Methodology, Writing – review & editing. **Yixuan Yuan:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by National Natural Science Foundation of China (62001410), Hong Kong Research Grants Council (RGC) Early Career Scheme grant 21207420 (CityU 9048179), Hong Kong RGC Collaborative Research Fund grant C4063-18G (CityU 8739029), and the InnoHK program.

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